SAI VENKATA NAGENDRA ABHAY RAJ GURRAM

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PROFESSIONAL SUMMARY

Machine Learning graduate student with experience building end-to-end ML workflows, including data preprocessing, model training, evaluation, and integration into applications. Strong foundation in supervised and unsupervised learning, NLP, and LLM-based systems, with hands-on experience using Python, PyTorch, and scikit-learn. Skilled in developing reproducible ML pipelines, applying standard evaluation metrics, and working with structured and unstructured data to solve real-world problems.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning & AI:
Supervised Learning, Unsupervised Learning, Model Training, Model Evaluation (Accuracy, Precision, Recall, F1), Cross-Validation, Overfitting & Regularization

Deep Learning & NLP:
PyTorch, Transformers, BERT, Hugging Face, Tokenization, Embeddings, Text Classification

ML Workflows & Systems:
Data Preprocessing, Feature Engineering, Training Pipelines, Evaluation Pipelines, Reproducible ML Workflows, Model Integration into Applications

Data & Analytics:
Pandas, NumPy, Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing

Data Engineering:
ETL Pipelines, Data Processing, Data Transformation, SQL Joins

Tools & Software Engineering:
Git, Version Control, Debugging, Code Structuring, Modular Design

Visualization:
Tableau, Power BI, Matplotlib

Cloud & Deployment:
AWS (S3, SageMaker – basic), FastAPI, Streamlit

Responsible AI:
Awareness of model fairness, privacy, security, and ethical AI practices

PROJECT EXPERIENCE

NovaMind – AI Decision Intelligence Platform

- Built an end-to-end ML system integrating LLM reasoning, RAG pipelines, and external tools for decision generation
- Implemented data preprocessing pipelines and structured workflows for model input preparation
- Designed modular multi-agent architecture to support scalable ML workflows
- Evaluated outputs using consistency and relevance metrics across test scenarios
- Integrated model outputs into a Streamlit-based application for real-time interaction
- Ensured reproducibility through structured pipeline design and modular components

AI Multi-Cloud GPU Cost Optimization System

- Built data ingestion and preprocessing pipelines for multi-source cloud pricing datasets
- Developed baseline decision models to evaluate cost-performance trade-offs
- Applied structured evaluation logic to compare deployment strategies
- Designed API-based integration for real-time inference and recommendations
- Implemented modular pipeline for data processing, model evaluation, and output generation

Rationale Agent – ML-Based Incident Analysis System

- Built ML-driven system to analyze structured and unstructured data for root cause identification
- Implemented preprocessing and feature extraction for reasoning workflows
- Designed evaluation framework for comparing multiple hypotheses
- Integrated ML outputs into an application interface for decision support
- Structured system for reproducibility and modular workflow execution

Exo-Cortex Debugger – AI-Based Code Analysis Tool

- Built Python-based system to execute code and capture runtime errors
- Implemented structured parsing and analysis of error outputs
- Integrated LLM-based explanations into application workflows
- Designed modular architecture supporting debugging, analysis, and output generation

BERT-Based Sentiment Analysis System

- Fine-tuned transformer-based model for text classification tasks
- Performed data preprocessing, tokenization, and embedding generation
- Evaluated model performance using accuracy and F1-score metrics
- Built visualization dashboards to analyze sentiment outputs
- Integrated model into application for real-time predictions

Formula 1 Strategy Analysis using Clustering

- Applied K-Means clustering to identify patterns in race strategies and performance metrics
- Performed data preprocessing and feature selection across multi-variable datasets
- Evaluated clustering outputs to derive meaningful performance segments
- Generated insights supporting data-driven decision-making

PROFESSIONAL EXPERIENCE

Quality Assurance Intern
Laurus Labs, Visakhapatnam, India | Jan 2022 – Jun 2022

- Performed data validation and analysis across structured datasets in regulated environments
- Identified inconsistencies and supported corrective actions using data-driven approaches
- Collaborated with teams to ensure process compliance and data accuracy
- Applied structured workflows for documentation, validation, and reporting

EDUCATION

Stevens Institute of Technology
Master's in Business Analytics & Artificial Intelligence
Expected May 2026 | GPA: 3.51

GITAM Deemed University
Bachelor's in Pharmacy

CERTIFICATIONS

Apple Search Ads Certification
Bank of America Investment Banking Simulation